

Lab 1

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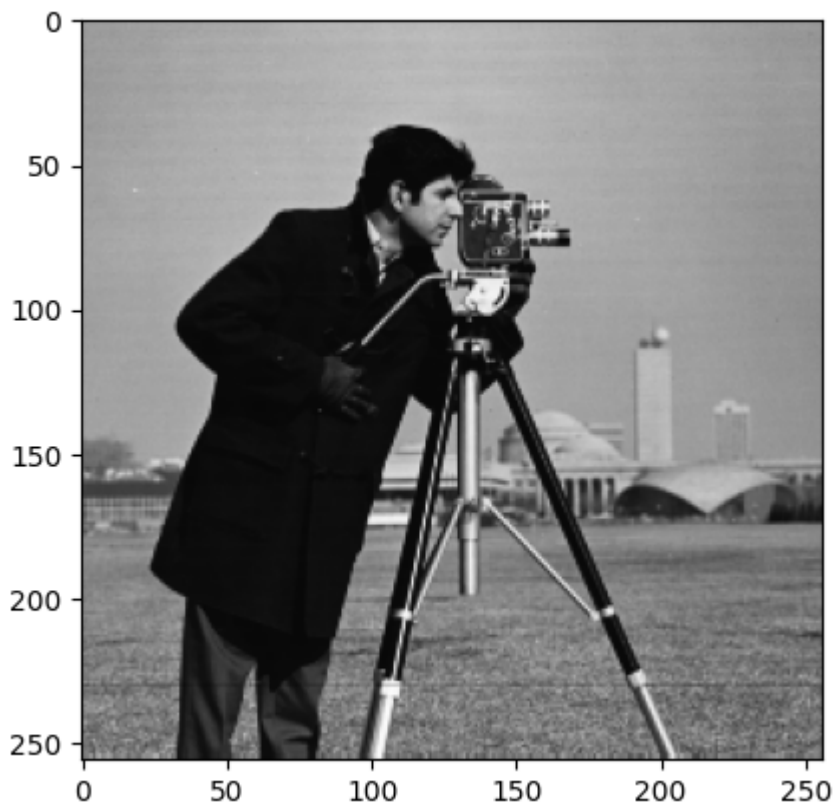
Group: 7MA1

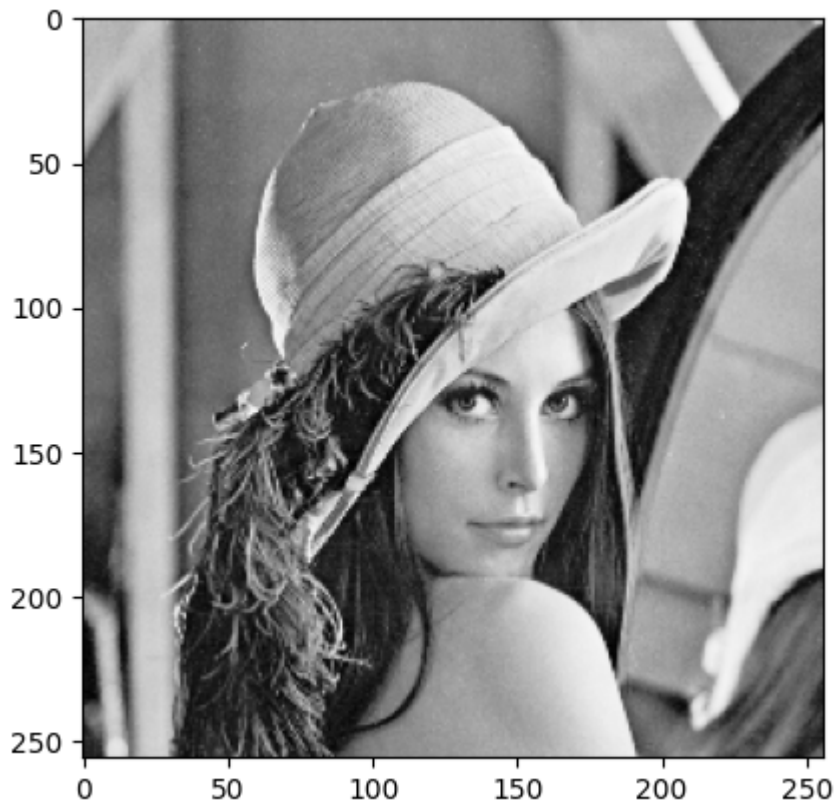
In [26]: *#Task2-a*

```
import cv2
img = cv2.imread('cameraman.tif') #Load the image data into a NumPy array

import matplotlib.pyplot as plt
plt.imshow(img)
plt.show()

#task2-b
from PIL import Image
img2 = Image.open('lena_gray_256.tif') #load image into a PIL Image object
import matplotlib.pyplot as plt
import matplotlib.cm as cm
plt.imshow(img2, cmap = cm.Greys_r)
plt.show()
```





```
In [27]: #Task3-a
cv2.imwrite('new_image.jpg', img)
#Task2-b
img2.save('new_image2.jpg')
```

```
In [28]: #Task4
print(img.shape) #Note: img1 loaded using open cv is a numpy array
print(img)

import numpy as np
# Convert the image to a NumPy array
img_array = np.array(img2)
# Print the array
print(img_array.shape)
print(img_array)
```

```
(256, 256, 3)
[[[156 156 156]
  [159 159 159]
  [158 158 158]
  ...
  [151 151 151]
  [152 152 152]
  [152 152 152]]

 [[160 160 160]
  [154 154 154]
  [157 157 157]
  ...
  [154 154 154]
  [155 155 155]
  [153 153 153]]

 [[156 156 156]
  [159 159 159]
  [158 158 158]
  ...
  [151 151 151]
  [152 152 152]
  [152 152 152]]

 ...

 [[114 114 114]
  [132 132 132]
  [123 123 123]
  ...
  [135 135 135]
  [137 137 137]
  [114 114 114]]

 [[121 121 121]
  [126 126 126]
  [130 130 130]
  ...
  [133 133 133]
  [130 130 130]
  [113 113 113]]
```

```
[[121 121 121]
 [126 126 126]
 [130 130 130]
 ...
 [133 133 133]
 [130 130 130]
 [113 113 113]]]
(256, 256)
[[162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 ...
 [ 48  57  54 ...  75  97  88]
 [ 46  50  52 ...  89 100  98]
 [ 44  55  54 ...  97 100 105]]
```